9.3 A RESONANT SWITCH CONVERTER: ZERO-VOLTAGE SWITCHING

Basic Operation

The circuit of Fig. 9-2*a* shows a method for using the oscillations of an *LC* circuit for switching at zero voltage. The analysis assumes that the output filter produces a ripple-free current I_o in L_o. Beginning with the switch closed, the current in the switch and in L_r is I_o, the currents in D_1 and D_s are zero, and the voltage across C_r and the switch is zero.

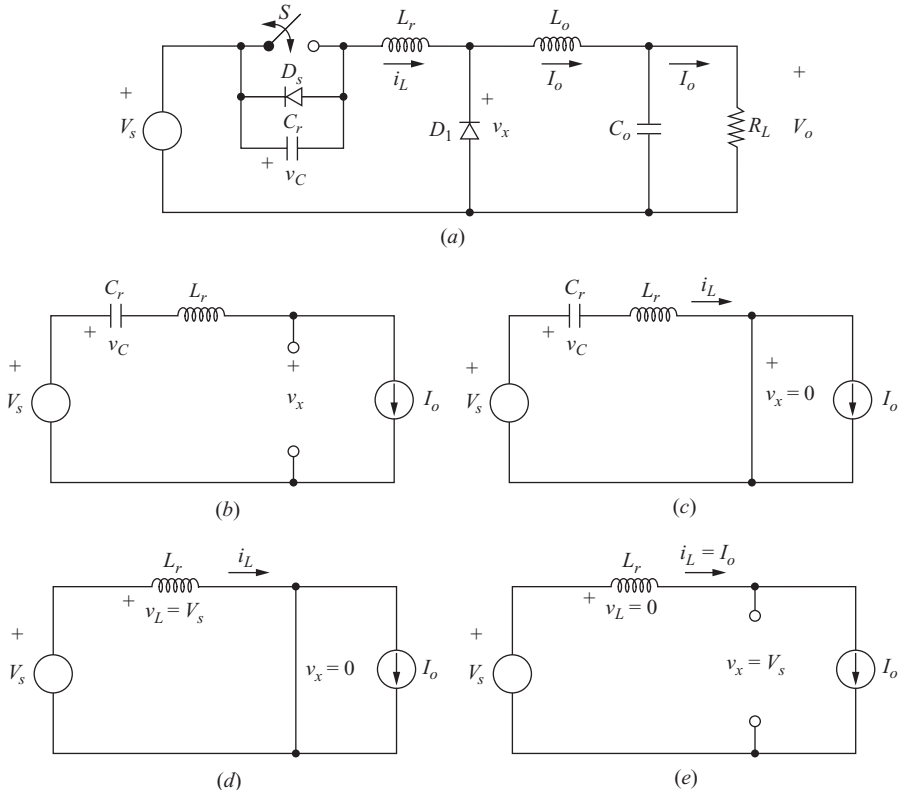
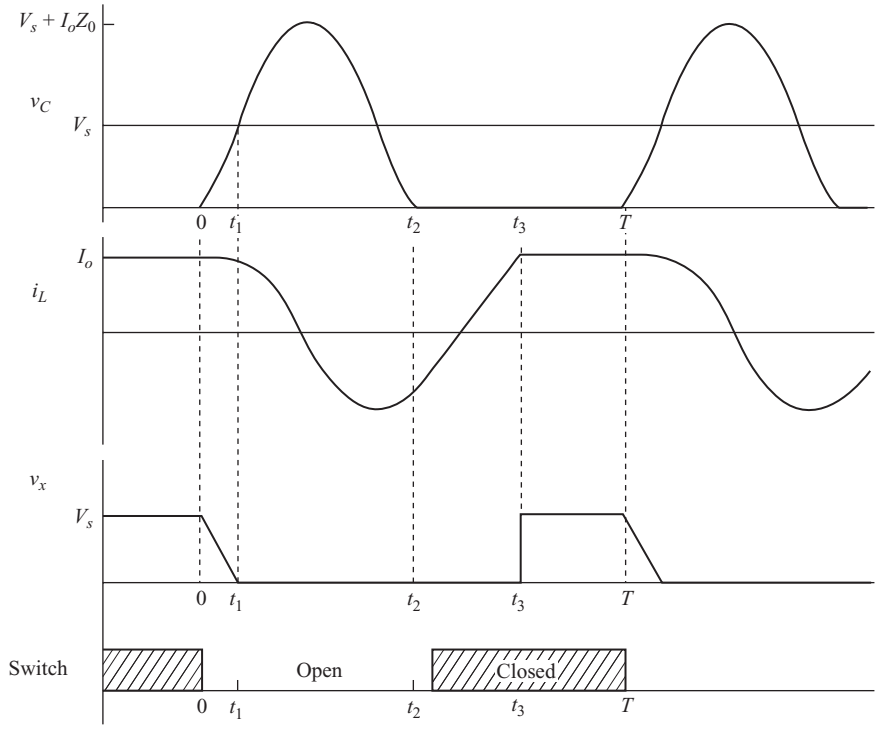


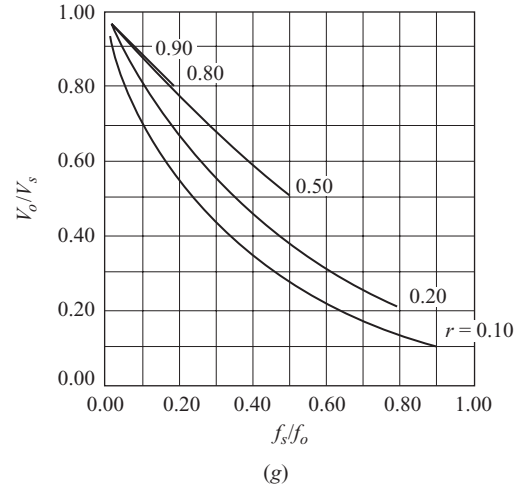
Figure 9-2 (*a*) A resonant converter with zero-voltage switching; (*b*) Switch open and D_1 off ($0 < t < t_1$); (*c*) Switch open and D_1 on ($t_1 < t < t_2$). (*d*) Switch closed and D_1 on ($t_2 < t < t_3$); (*e*) Switch closed and D_1 off ($t_3 < t < T$); (*f*) Waveforms (*g*) Normalized output vs. switching frequency with $r = R_L/Z_0$ as a parameter. © 1992 IEEE, B.K. Bose, Modern Power Electronics: Evolution, Technology, and Applications. Reprinted with permission.

The switch is opened (with zero voltage across it), and $i_L = I_o$ flows through the capacitor C_r, causing v_C to increase linearly (Fig. 9-2*b*). When v_C reaches the source voltage V_s, the diode D_1 becomes forward-biased, in effect forming a series circuit with V_s, C_r, and L_r as shown in Fig. 9-2*c*. At this time, i_L and v_C in this underdamped series circuit begin to oscillate.

When v_C returns to zero, diode D_s turns on to carry i_L, which is negative (Fig. 9-2*d*). The voltage across L_r is V_s, causing i_L to increase linearly. The switch should be closed just after D_s turns on for zero-voltage turn-on. When i_L becomes positive, D_s turns off and i_L is carried by the switch. When i_L reaches I_o, D_1 turns off, and circuit conditions are back at the starting point. The analysis for each circuit condition is given next.



(*f*)



(*g*)

Figure 9-2 (*continued*)

Analysis for $0 \le t \le t_1$ The switch is opened at $t = 0$. The capacitor current is then I_o (Fig. 9-2*b*), causing the capacitor voltage, initially zero, to increase linearly. The voltage across C_r is

$$v_C(t) = \frac{1}{C_r}\int_0^t I_o \, d\lambda = \frac{I_o}{C_r}t \tag{9-22}$$

The voltage across L_r is zero because inductor current is I_o, which is assumed to be constant. The voltage at the filter input v_x is

$$v_x(t) = V_s - v_C(t) = V_s - \frac{I_o}{C_r}t \tag{9-23}$$

which is a linearly decreasing function beginning at V_s. At $t = t_1$, $v_x = 0$ and the diode turns on. Solving the preceding equation for t_1,

$$\boxed{t_1 = \frac{V_s C_r}{I_o}} \tag{9-24}$$

Equation (9-23) can then be expressed as

$$v_x(t) = V_s\left(1 - \frac{t}{t_1}\right) \tag{9-25}$$

Analysis for $t_1 \le t \le t_2$ Diode D_1 is forward-biased and has 0 V across it, and the equivalent circuit is shown in Fig. 9-2*c*. Kirchhoff's voltage law is expressed as

$$L_r\frac{di_L(t)}{dt} + v_C(t) = V_s \tag{9-26}$$

Differentiating,

$$L_r\frac{d^2i_L(t)}{dt^2} + \frac{dv_C(t)}{dt} = 0 \tag{9-27}$$

Capacitor current is related to voltage by

$$\frac{dv_C(t)}{dt} = \frac{i_C(t)}{C_r} \tag{9-28}$$

Since inductor and capacitor currents are the same in this time interval, Eq. (9-27) can be expressed as

$$\frac{d^2i_L(t)}{dt^2} + \frac{i_L(t)}{L_rC_r} = 0 \tag{9-29}$$

Solving the preceding equation for i_L by using the initial condition $i_L(t_1) = I_o$,

$$i_L(t) = I_o \cos[\omega_0(t - t_1)] \tag{9-30}$$

where

$$\omega_0 = \frac{1}{\sqrt{L_r C_r}} \tag{9-31}$$

Capacitor voltage is expressed as

$$v_C(t) = \frac{1}{C_r}\int_{t_1}^{t} i_C(\lambda)\, d\lambda + v_C(t_1) = \frac{1}{C_r}\int_{t_1}^{t} I_o \cos[\omega_0(\lambda - t_1)]\, d\lambda + V_s$$

which simplifies to

$$v_C(t) = V_s + I_o Z_0 \sin[\omega_0(t - t_1)] \tag{9-32}$$

where

$$Z_0 = \sqrt{\frac{L_r}{C_r}} \tag{9-33}$$

Note that the peak capacitor voltage is

$$V_{C,\text{peak}} = V_s + I_o Z_0 = V_s + I_o \sqrt{\frac{L_r}{C_r}} \tag{9-34}$$

which is also the maximum reverse voltage across diode D_s and is larger than the source voltage.

With diode D_1 forward-biased,

$$v_x = 0 \tag{9-35}$$

The diode D_s across C_r prevents v_C from going negative, so Eq. (9-32) is valid for $v_C > 0$. Solving Eq. (9-32) for the time $t = t_2$ when v_C returns to zero,

$$t_2 = \frac{1}{\omega_0}\left[\sin^{-1}\left(\frac{-V_s}{I_o Z_0}\right)\right] + t_1$$

which can be expressed as

$$\boxed{t_2 = \frac{1}{\omega_0}\left[\sin^{-1}\left(\frac{V_s}{I_o Z_0}\right) + \pi\right] + t_1} \tag{9-36}$$

At $t = t_2$, diode D_s turns on.

Analysis for $t_2 \le t \le t_3$ (Fig. 9-2*d*) After t_2, both diodes are forward-biased (Fig. 9-2*d*), the voltage across L_r is V_s, and i_L increases linearly until it reaches I_o at t_3. The switch is reclosed just after t_2 when $v_C = 0$ (zero-voltage turn-on) and the diode is on to carry a negative i_L. The current i_L in the interval from t_2 to t_3 is expressed as

$$i_L(t) = \frac{1}{L_r}\int_{t_2}^{t} V_s\, d\lambda + i_L(t_2) = \frac{V_s}{L_r}(t - t_2) + I_o \cos[\omega_0(t_2 - t_1)] \tag{9-37}$$

where $i_L(t_2)$ is from Eq. (9-30). Current at t_3 is I_o:

$$i_L(t_3) = I_o = \frac{V_s}{L_r}(t_3 - t_2) + I_o \cos[\omega_0(t_2 - t_1)] \tag{9-38}$$

Solving for t_3,

$$\boxed{t_3 = \left(\frac{L_r I_o}{V_s}\right)\{1 - \cos[\omega_0(t_2 - t_1)]\} + t_2} \tag{9-39}$$

Voltage v_x is zero in this interval:

$$v_x = 0 \tag{9-40}$$

At $t = t_3$, diode D_1 turns on.

Analysis for $t_3 \le t \le T$ In this interval, the switch is closed, both diodes are off, the current in the switch is I_o, and

$$v_x = V_s \tag{9-41}$$

The circuit remains in this condition until the switch is reopened. The time interval $T - t_3$ is determined by the switching frequency of the circuit. All other time intervals are determined by other circuit parameters.

Output Voltage

The voltage $v_x(t)$ at the input of the output filter is shown in Fig. 9-2*f*. Summarizing Eqs. (9-25), (9-35), (9-40), and (9-41),

$$v_x(t) = \begin{cases} V_s\left(1 - \dfrac{t}{t_1}\right) & 0 < t < t_1 \\ 0 & t_1 < t < t_3 \\ V_s & t_3 < t < T \end{cases} \tag{9-42}$$

The output voltage is the average of $v_x(t)$. Output voltage is

$$V_o = \frac{1}{T}\int_0^T v_x\,dt = \frac{1}{T}\left[\int_0^{t_1} V_s\left(1 - \frac{t}{t_1}\right)dt + \int_{t_3}^{T} V_s\,dt\right] \tag{9-43}$$
$$= \frac{V_s}{T}\left[\frac{t_1}{2} + (T - t_3)\right]$$

Using $f_s = 1/T$,

$$\boxed{V_o = V_s\left[1 - f_s\left(t_3 - \frac{t_1}{2}\right)\right]} \tag{9-44}$$

Times t_1, and t_3 in the preceding equation are determined from the circuit parameters as described by Eqs. (9-24), (9-36), and (9-39). *The output voltage is controlled by changing the switching frequency*. The time interval when the switch is open is fixed, and the time interval when the switch is closed is varied. Times t_1 and t_3 are determined in part by the load current I_o, so output voltage is a function of load. Increasing the switching frequency decreases the time interval $T - t_3$ and thus reduces the output voltage. Normalized output voltage vs. switching frequency with the parameter $r = R_L/Z_0$ is shown in the graph in Fig. 9-2*g*. Output voltage is less than input voltage, as was the case for the buck converter in Chap. 6.

EXAMPLE 9-2

Resonant Switch Converter: Zero-Voltage Switching

In the circuit of Fig. 9-2*a*,

$$V_s = 20\text{ V}$$
$$L_r = 1\ \mu\text{H}$$
$$C_r = 0.047\ \mu\text{F}$$
$$I_o = 5\text{ A}$$

(*a*) Determine the switching frequency such that the output voltage is 10 V.
(*b*) Determine the peak voltage across D_s when it is reverse-biased.

■ Solution

(*a*) From the circuit parameters,

$$\omega_0 = \frac{1}{\sqrt{(10)^{-6}(0.047)(10)^{-6}}} = 4.61(10^6)\text{ rad/s}$$

$$Z_0 = \sqrt{\frac{L_r}{C_r}} = \sqrt{\frac{10^{-6}}{0.047(10^{-6})}} = 4.61\ \Omega$$

Using Eq. (9-24) to solve for t_1,

$$t_1 = \frac{V_s C_r}{I_o} = \frac{(20)(0.047)(10^{-6})}{5} = 0.188\ \mu\text{s}$$

From Eq. (9-36),

$$t_2 = \frac{1}{\omega_0}\left[\sin^{-1}\left(\frac{V_s}{I_o\sqrt{L_r/C_r}}\right) + \pi\right] + t_1$$

$$= \frac{1}{4.61(10^6)}\left[\sin^{-1}\frac{20}{(5)(4.61)} + \pi\right] + 0.188\ \mu\text{s} = 1.10\ \mu\text{s}$$

From Eq. (9-39),

$$t_3 = \left(\frac{L_r I_o}{V_s}\right)\{1 - \cos[\omega_0 (t_2 - t_1)]\} + t_2$$

$$= \left(\frac{10^{-6}(5)}{20}\right)\{1 - \cos[(4.61)(10^6)(1.10 - 0.188)(10^{-6})]\} + 1.10\ \mu\text{s} = 1.47\ \mu\text{s}$$

Equation (9-44) is used to determine the proper switching frequency,

$$V_o = V_s\left[1 - f_s\left(t_3 - \frac{t_1}{2}\right)\right]$$

$$10 = 20\left[1 - f_s\left(1.47 - \frac{0.188}{2}\right)(10^{-6})\right]$$

$$f_s = 363\ \text{kHz}$$

(*b*) Peak reverse voltage across D_s is the same as peak capacitor voltage. From Eq. (9-25),

$$V_{D_s,\text{peak}} = V_{C,\text{peak}} = V_o + I_o\sqrt{\frac{L_r}{C_r}} = 20 + (5)(4.61) = 33\ \text{V}$$